INCH-POUND

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29 April 2025
SUPERSEDING
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DETAIL SPECIFICATION

METHYL CENTRALITE (FOR AMMUNITION USE)

Reactivated after 27 January 2025 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two classes of methyl centralite for use in the manufacture of propellants.

1.2 Classification. Methyl centralite is of the following Classes (see 6.2):

Class 1 – Lump
Class 2 – Powder

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army DEVCOM Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at (https://quicksearch.dla.mil/.)

2.3 Non-Government standards and publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1193	-	Standard Specification for Reagent Water
ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E300	-	Standard Practice for Sampling Industrial Chemicals
ASTM E1252	-	Standard Practice for General Techniques for Obtaining Infrared Spectra for Qualitative Analysis

(Copies of these documents are available from https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

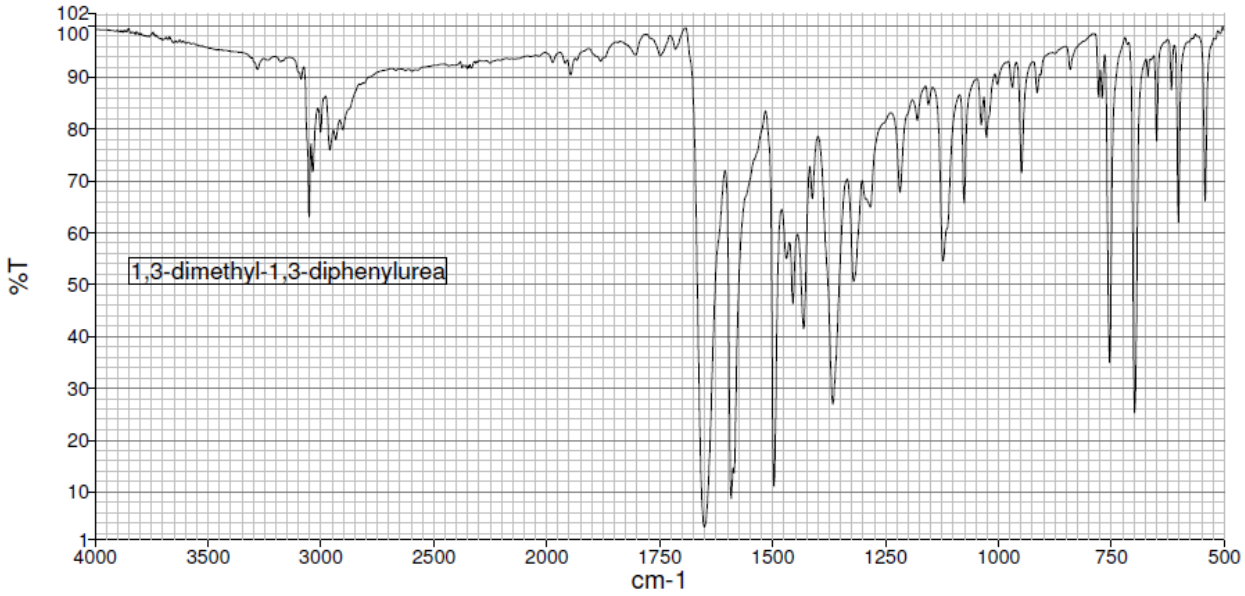
3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Material infrared spectrum. Methyl centralite shall consist of essentially 1,3-dimethyl-1,3-diphenylurea urea [$CO(NCH_3C_6H_5)_2$] whose infrared (IR) spectrum shall contain the major peaks shown in Figure 1 and shall contain no large extraneous peaks.

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Major peaks, cm^{-1}: 3054, 1651, 1591, 1585, 1497, 1469, 1455, 1431, 1366, 1320, 1283, 1122, 1075, 752, 698

FIGURE 1. IR spectrum for methyl centralite.

3.3 Appearance of melted material. Methyl centralite shall melt to a bright clear liquid, free from scum and deposit.

3.4 Chemical and physical requirements. Methyl centralite shall conform to the chemical and physical characteristics of Table I when tested as specified therein.

TABLE I. Chemical and physical requirements.

Characteristic	Requirement
Solidification point or Melting point characteristics……Onset of melt Clear-point Onset to clear-point Sinter to onset point	120.5 – 122.0˚C 121.1˚C minimum (min.) 122.5 – 124.0˚C 1.8˚C maximum (max.) 0.5˚C max.
Volatile matter (at 100˚C)	0.20 (weight percent, wt.%) max.
Ash content	0.1 wt.% max.
Hydrolyzable chlorine compounds	No indication
Acidity (as acetic acid)	0.006 wt.% max.
Amines content	0.20 wt.% max.
Granulation (Class 2 only) Thru U.S. Alternative No. 50 (see 1/)	 99.55 wt.% min. (see 6.9)
Notes: 1/ Sieves shall be in accordance with ASTM E11.	

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3.5 Workmanship. Methyl centralite shall be a white crystalline, lump or powder, material which is free from grit, visible impurities, and foreign material such as dirt, rust, paint, wooden particles or any other foreign objects.

4. VERIFICATION

TABLE II. Requirements/verification cross reference matrix.

Methods of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article	X		X		X		4.2
3.1.2	Conformance	X		X			X	4.3
3.2	Material infrared spectrum			X		X	X	4.4.1
3.3	Solidification point / melting point	X				X	X	4.4.2
3.4	Appearance of melted material	X				X	X	4.4.3
3.4	Volatile matter	X				X	X	4.4.4
3.4	Ash content	X				X	X	4.4.5
3.4	Hydrolyzable chlorine compounds	X				X	X	4.4.6
3.4	Acidity (as acetic acid)	X				X	X	4.4.7
3.4	Amines content	X				X	X	4.4.8
3.4	Granulation (Class 2)	X				X	X	4.4.9
3.5	Workmanship			X		X	X	4.4.10

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2).
b. Conformance inspection (see paragraph 4.3).

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table III. Samples collected in accordance with 4.3.2 shall be submitted for examination. Samples shall be produced using the same process procedures, material supply sources, production equipment, and production line as will be used in regular production.

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TABLE III. First article acceptance tests.

Category	Examination or test	Requirement paragraph	Inspection method
Critical	None defined		
Major			
101	Material infrared spectrum	3.2	4.4.1
102	Solidification point / melting point	3.3	4.4.2
103	Appearance of melted material	3.4	4.4.3
104	Volatile matter	3.4	4.4.4
105	Ash content	3.4	4.4.5
106	Hydrolyzable chlorine compounds	3.4	4.4.6
107	Acidity (as acetic acid)	3.4	4.4.7
108	Amines content	3.4	4.4.8
109	Granulation (Class 2)	3.4	4.4.9
Minor			
201	Workmanship	3.5	4.4.10

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. Inspection lots and batches shall comply with the MIL-STD-1916 paragraph "Formation and identification of lots or batches." Each batch shall consist of that quantity of methyl centralite that has been subjected to the same chemical and physical mixing process intended to make the final product homogenous. A lot shall consist of one or more acceptable batches of methyl centralite produced by one manufacturer in accordance with one specification and revision under one continuous set of operating conditions. A lot shall consist of the methyl centralite produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.3.2 Methyl centralite sampling. Sampling for inspection shall be performed at either the supplier/distributor location where the methyl centralite is packed, or at the manufacturer's receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation.

4.3.2.1 Sampling at the supplier/distributor. Samples shall be collected from the beginning, middle, and end of every batch in every lot in accordance with ASTM E300. There shall be no composite sampling. All samples shall be verified in accordance with 4.4. If any sample fails to comply with any applicable requirements the batch shall be rejected.

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4.3.2.2 Sampling at the receiving facility. Methyl centralite packaged in shipping containers (e.g., drums, boxes, etc.) shall be sampled using Verification Level I of MIL-STD-1916. Each shipping container shall be sampled in three locations in accordance with ASTM E300 using a thief stick. The representative specimen from each shipping container shall be not less than 450 g and shall be placed in a suitable clean, dry container labeled to identify the lot and shipping container from which it was taken. There shall be no composite sampling. There shall be no reduced sampling. Each sample from every lot received shall be tested in accordance with 4.4. If any sample fails to meet any test requirement, the lot represented by the sample shall be rejected.

4.3.3 Classification of characteristics.

4.3.3.1 Methyl centralite.

Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Material infrared spectrum	4.3.2	3.2	4.4.1
102	Solidification point / melting point	4.3.2	3.3	4.4.2
103	Appearance of melted material	4.3.2	3.4	4.4.3
104	Volatile matter	4.3.2	3.4	4.4.4
105	Ash content	4.3.2	3.4	4.4.5
106	Hydrolyzable chlorine compounds	4.3.2	3.4	4.4.6
107	Acidity (as HNO_3)	4.3.2	3.4	4.4.7
108	Amines content	4.3.2	3.4	4.4.8
109	Granulation (Class 2)	4.3.2	3.4	4.4.9
Minor				
201	Workmanship	4.3.2	3.5	4.4.10

4.4 Test methods and procedures. Water in accordance with ASTM D1193, Type I or II, and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. If necessary, Class 1 material shall be ground to pass through a U.S. Alternative No. 60 sieve. Tests shall be conducted as follows:

4.4.1 Material infrared spectrum. Verify by examination that an infrared spectrum collected in accordance with ASTM E1252 matches a reference standard of methyl centralite and contains the major peaks identified in Figure 1.

4.4.1.1 Equipment and reagents.

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a. Fourier-transform infrared spectrophotometer (FTIR), set up for transmission mode
b. Potassium bromide (KBr) saltplate
c. Acetone, reagent grade

4.4.1.2 Procedure. Use the KBr saltplate to blank the FTIR: collect and average ten (10) spectra in the 4000-500 cm-1 range at 4 cm-1 resolution. Prepare the test solution by dissolving 20 mg of the sample in 2 mL of dry acetone. Spread a few microliters of the solution onto the KBr saltplate and allow to air dry. Scan the sample using the same parameters as the blank, whose spectrum shall be automatically subtracted from the sample. Convert the resulting spectrum to percent transmittance and compare with Figure 1.

4.4.2 Solidification point / melting point. Determine solidification point or melting in accordance with the following methods as appropriate.

4.4.2.1 Solidification point determination. Place about 50 grams of the sample in a glass beaker, cover the beaker, and dry in an oven at 100˚C ± 5˚C overnight or in a desiccator for 24 hours. Use the dried sample to fill the inner tube of the solidification point apparatus (see Figure 2) to within 1½ inch of the top. Loosely cap the test tube with a cork, place the apparatus with sample in an oven or glycerin bath, and heat until sample is molten (see 4.4.3). Close the test tube with the cork carrying the stirrer and calibrated, total immersion thermometer. Adjust the height of the thermometer so that the bulb is in the center of the molten sample and note the temperature marking at which the stem emerges from the melt. Place the tube in its place in the apparatus. Stir slowly and constantly, reading the temperature every 15 seconds. Continue stirring until the temperature stays steady for three readings after the rise from super cooling. Carefully watch the mercury thread and note the maximum temperature reached and the length of time this maximum is maintained. Determine the correction for emergent stem by determining the average temperature of the exposed mercury column of the standard thermometer by means of a second thermometer suspended so that its bulb is adjacent to about the middle of the exposed mercury column of the standard thermometer.

$$\text{Emergent stem correction} = (T - m) \times (T - t) \times (0.000159)$$

Where: T = the uncorrected solidification point, i.e. maximum temperature reached after rise from supercooling, ˚C
m = temperature marking on standard thermometer touching the surface of the melt, ˚C
t = average temperature of the exposed mercury column determined by means of a second thermometer suspended so that its bulb is at (T+m)/2, ˚C
0.000159 = the coefficient of expansion of mercury in glass, 1/˚C

Calculate and record the corrected reading as the solidification point of the sample. Corrected reading in ˚C is T + emergent stem correction.

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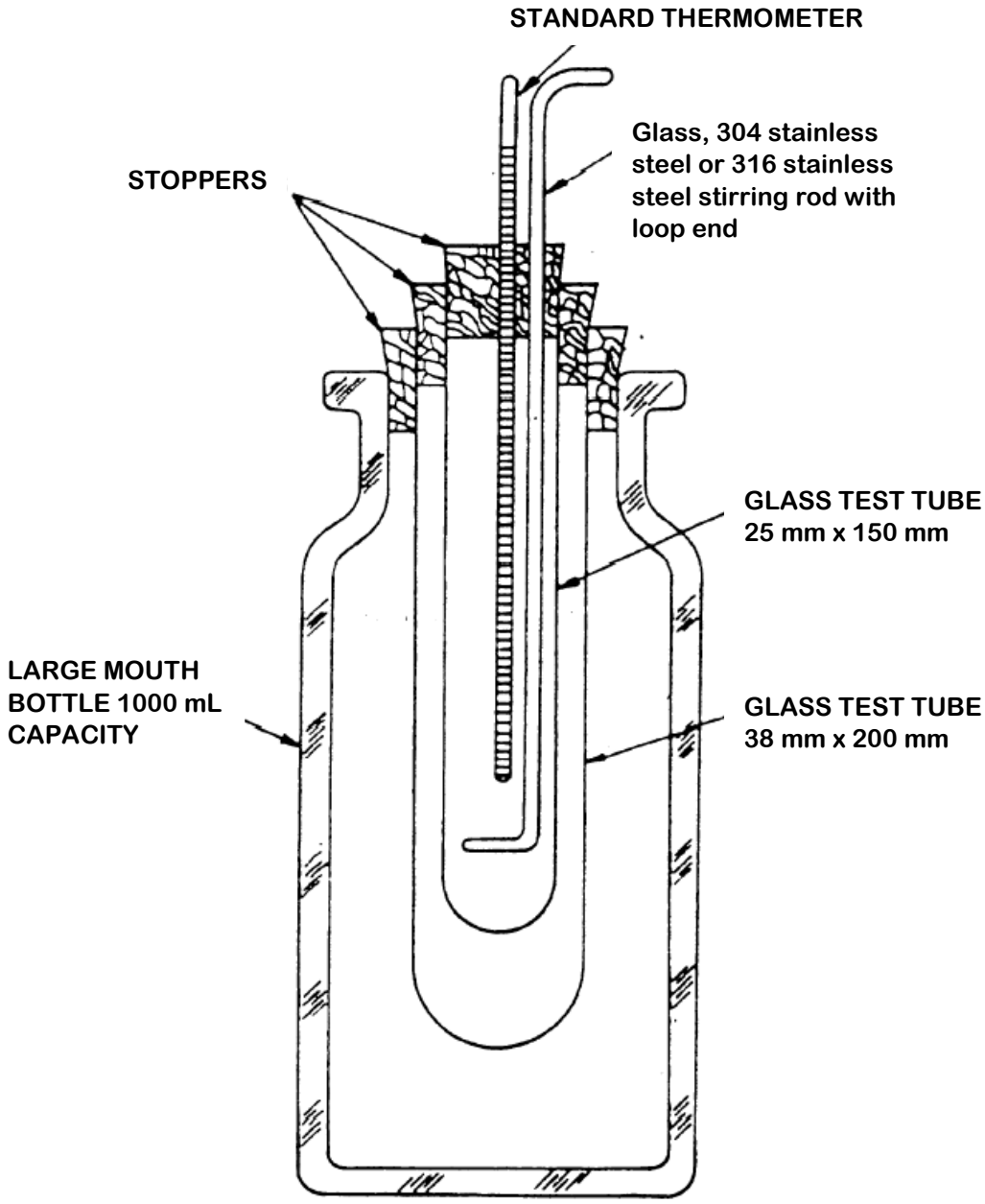


FIGURE 2. Apparatus for determining solidification or setting point.

4.4.2.2 Melting point determination (alternate).

4.4.2.2.1 Reagents for melting point determination.

a. Benzoic acid, analytical standard intended for the calibration of melting point system, powdered and dried in desiccator for at least 16 hours.

4.4.2.2.2 Equipment.

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a. Melting point system conforming to USP Section 741, Apparatus I or II (see 6.11), wherein the sample may be observed directly or through the use of video.

b. Capillary tubes about 0.8-1.2 mm internal diameter with 0.2-0.3 mm wall thickness, 7-10 cm long, closed at one end; or as required by the apparatus.

4.4.2.2.3 Calibration.

a. Fill three (3) capillary tubes with enough benzoic acid to form a 2.5-3.5 mm column on the bottom of the tube by dropping the tubes vertically onto a solid surface several times from a height of a few centimeters.

b. Preheat the melting point apparatus to 118˚C and insert the tubes into its heating block. Continue heating at 1˚C/min.

c. Record the average clear point temperature, compare it with the melting point specified for benzoic acid by the manufacturer - (typically 124.0 ± 0.5˚C), and correct all subsequent temperature readings by the discrepancy.

4.4.2.2.4 Analysis.

a. For Class 1 material, gently grind one (1) gram in a mortar prior to drying. For either Class, dry the material in a desiccator for at least 16 hours.

b. Fill three (3) capillary tubes with enough sample to form a 2.5-3.5 mm column on the bottom of the tube by dropping the tubes vertically onto a solid surface several times from a height of a few centimeters.

c. Preheat the melting point apparatus to 118˚C and insert the tubes into its heating block. Continue heating at 1˚C/min. For each tube, record the temperature at the first sign of sintering, when crystals adjacent to the capillary wall begin to melt.

d. Record the onset temperature when the solid column begins to reduce in height and collapse.

e. Record the first clear-point temperature when only the liquid phase remains.

f. Calculate the mean average temperature for the three tubes at the start of sintering, at onset, and at minimum clear point; record the latter two averages for conformance.

g. Calculate and record the difference between the average sinter start and onset, and the difference between average onset and clear-point for conformance.

4.4.3 Appearance of melted material. Appearance of melted material may be visually verified during the solidification point test (4.4.2.1). Alternately, place about 50 grams of the sample in a glass beaker, cover the beaker, and dry in an oven at 100˚C ± 5˚C overnight or in a desiccator for 24 hours. Then melt the sample in an oven at 125-130˚C and visually examine for appearance of melted material.

4.4.4 Volatile matter. Transfer a 10 gram portion of the sample, weighed to the nearest 0.1 milligram, to a tared, dry, glass-covered container, and record the gross weight to the nearest 0.1 mg. Loosely stopper the container and heat in an oven at 100˚C to 105˚C for three hours, cool in a desiccator, and weigh. Calculate the loss in weight as percentage of volatile matter as follows:

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$$\text{Percent of volatile matter} = \frac{\text{Loss in weight}}{\text{Weight of sample}} \times 100$$

4.4.5 Ash content. Prepare a porcelain crucible by cleaning with ethanol and burning it out, allowing to cool to ambient, and storing in a desiccator prior to use. Transfer an accurately weighed sample of approximately 5 grams of the material to the tared porcelain crucible and add sufficient 95 percent ethyl alcohol to cover the material. Ignite the mixture and allow it to burn down slowly and then heat to redness until all carbon has been removed. Calculate the increase in weight to percentage ash in the sample as follows:

$$\text{Percent ash} = \frac{\text{Increase in weight of tared crucible}}{\text{Weight of sample}} \times 100$$

4.4.6 Hydrolyzable chlorine compounds.

4.4.6.1 Equipment and supplies.

a. Analytical balance with 200 g capacity and 0.1 mg resolution
b. Conical flasks, 125 mL or 250 mL beakers
c. pH test paper, universal
d. hot plate
e. Whatman No. 40 filter paper
f. Nessler tubes, 50 mL (approximately 20 cm viewing length, 1.7 cm inner diameter)

4.4.6.2 Stock solutions and reagents for chlorine compound determination. All chemicals used shall be reagent grade, or as specified below

a. Water, deionized (DIW) or distilled, conforming to ASTM D1193 Type I or II.
b. Sodium ethoxide, 5% v/v in ethanol.
c. Silver nitrate, 10 %w/v: dissolve 10 g of silver nitrate in water and dilute to 100 mL. Acidify with concentrated nitric acid to pH 1.

4.4.6.3 Procedure. Each sample is analyzed using two (2) replicates. A reagent blank shall be prepared as well.

a. Weigh 5.0 g of methyl centralite and dissolve in 10 mL of a 5%wt sodium ethoxide in ethanol by heating atop a steam bath or hot plate (~130˚C surface temperature) until the sample completely dissolves.
b. Dilute to 100 mL with DI water, mix well, cool to room temperature and allow the bulk of methyl centralite to crystallize.
c. Filter the solution through a Whatman #40 filter which has been twice pre-washed with hot DIW.
d. Acidify the filtrate with 1 mL of concentrated nitric acid.
e. Cover and leave the filtrate for 1-2 hours to allow the remaining methyl centralite to crystallize out.

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f. Refilter again through Whatman #40 filter which has been twice pre-washed with hot DI water.

g. Add 5 mL of 10% w/v silver nitrate solution with continuous stirring.

h. Let the solutions stand in the dark for not less than thirty minutes.

i. Transfer 50 mL of each solution to separate Nessler tubes.

j. Check turbidity of the samples by looking lengthwise through the tubes against a dark background.

k. The turbidity (opacity) of the sample solutions shall be not greater than that produced by the reagent blank containing equal quantities of all reagents except sample.

4.4.7 Acidity. Dissolve 20 grams of methyl centralite in 100 mL of hot toluene in a conical flask. Stopper, allow to cool to room temperature and for the excess sample to crystallize out. Transfer the supernatant to a separatory funnel. Rinse the residue with 5-10 mL toluene and combine with bulk solution already in the funnel. Extract with three successive 30 mL portions of freshly boiled and cooled distilled or deionized water. To the combined extract add 3 drops of a 1% tincture of phenolphthalein as indicator and titrate with 0.02 N sodium hydroxide solution to a light-pink endpoint. Carry out a blank estimation concurrently with the sample determination. Calculate acidity as follows:

$$\text{Acidity, as percent acetic acid} = \frac{6.0 \times N \times (V - B)}{W}$$

Where: N = normality of sodium hydroxide solution
V = volume of sodium hydroxide solution for sample
B = volume of sodium hydroxide solution for blank
W = weight of sample

4.4.8 Amines content. Determine amine content in accordance with the following procedure.

4.4.8.1 Equipment and supplies.

a. Analytical balance with 200 gram capacity and 0.1 mg resolution
b. Conical flasks, 125 mL and 500 mL
c. Hot plate
d. Glass stirring rod
e. Separatory funnel, 250 mL
f. Volumetric pipettes, Class A (see 6.10), 5 mL and 25 mL
g. Burette, Class A, 50 mL

4.4.8.2 Reagents for amines content determination. All chemicals shall be reagent grade or as specified.

a. Water, deionized (DI) or distilled
b. Toluene
c. Acetic anhydride

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d. Acetic anhydride, 0.25M solution in toluene; prepare by diluting 1.32 ± 0.05 g acetic anhydride in 50 mL toluene
e. Sodium hydroxide, 0.1N standardized solution
f. Hydrochloric acid, 0.1N standardized solution
g. Phenolphthalein, 1 wt.% solution in ethanol

4.4.8.3 Procedure. This method determines total primary and secondary amines, and other organic groups such as hydroxyl susceptible to acetylation. A correction is made for sample acidity. Each sample shall be analyzed using two separate aliquots of methyl centralite. Two blanks shall be prepared to ensure reproducibility.

a. Dissolve 25.0 g of sample in 30 mL hot (approximately 100 – 110˚C) toluene in a 125 mL conical flask. For the blanks, heat 30 mL toluene in a 125 mL flask alongside the sample.
b. Remove the solutions from heat, add 5.0 mL of 0.25M acetic anhydride solution in toluene, and mix well. Stopper the flasks and leave them overnight.
c. Decant the supernatant toluene quantitatively into the separatory funnel while keeping the precipitate in the flask. Do not allow the precipitate to pass to the funnel in large quantity as this will interfere with the titration.
d. Wash the crystals and flask with another 5-10 mL of toluene, mix well, and decant to the separatory funnel.
e. Add 25.0 mL of 0.1N sodium hydroxide to the flask, mix well, and decant to the separatory funnel. Shake the funnel thoroughly and let the phases separate.
f. Collect the lower layer in a 500 mL flask containing 25.0 mL of 0.1 N hydrochloric acid.
g. Wash the methyl centralite crystals in the 125 mL flask with 25 mL DI water, then decant the wash solution to the separatory funnel.
h. Shake the funnel thoroughly, then add the lower layer to the collected aqueous phase in the 500 mL flask.
i. Repeat rinsing the flask and funnel with DI water an additional four (4) times. The same washing procedure is prescribed for the blank.
j. Titrate with 0.1N sodium hydroxide solution using phenolphthalein as indicator.
k. Calculate amine content as follows:

$$\text{Percent of methylaniline} = \frac{10.7 \times (B - V)}{W} - (A \times 1.78)$$

Where: B = volume of sodium hydroxide required to titrate blank (average value for the two blanks), mL
V = volume of sodium hydroxide required to titrate sample
N = Normality of sodium hydroxide solution
W = weight of sample
A = percent acidity as acetic acid (see 4.4.7)

l. Results for the two replicates shall differ by no more than 0.02 %wt. Record the average for conformance.

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4.4.9 Granulation (Class 2). Fit a U.S. Alternative No. 50 sieve with a bottom pan. Place a weighed portion of 100 grams of the sample on the sieve. Cover and shake for 10 minutes by hand or for 5 minutes by means of a mechanical shaker geared to produce 268 – 315 gyrations and 150 ± 10 taps of the striker per minute. Weigh the amount retained on the sieve (if any) and calculate the percentage of material passing through the sieve.

4.4.10 Workmanship. Take approximately 100 grams of the material and spread the composition out on a clean white surface. Verify by visual examination that the sample is white, crystalline, and lump or powder as appropriate for the Class. Verify that the sample is uniform and free from visible impurities, foreign matter or other foreign objects. Failure to meet the workmanship inspection or during following production processes shall result in batch or lot rejection.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Methyl centralite is used as a coating agent in propellants and in the manufacture of propellant powders.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Packaging requirements (see 5.1).
c. Class of material.
d. Provisions for submission/resubmission of first article samples.
e. Quantity required and delivery schedules.
f. Requirements for submission of acceptance inspection equipment (see 6.7).
g. Certificate(s) of Analysis reporting the test results for the batch(es) or container(s) tested.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process of subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Technical agency. US Army Combat Capability Development Command - Armaments Center (DEVCOM AC) is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: DEVCOM AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Approval of equivalent test methods. Prior approval by the Contracting Officer should be required for use of equivalent test methods and/or instruments. A description of the proposed method and/or instrumentation should be submitted through the Contracting Officer to the Technical Agency (see 6.4). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.6 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency (See 6.4) responsible for preparation of this specification.

6.7 Submission of Acceptance Inspection Equipment (AIE). Provisions concerning the contractor's AIE used to verify the requirements of this specification should be specified in the contract. Address for submission should be specified on the Contract Data Requirement List, DD Form 1423 in the contract. Inspection equipment used for acceptance of product should be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% of the total tolerance of the characteristic being inspected. This inspection requirement should also be specified in the contract.

6.8 Water. Water used for acidity verification should be recently boiled and cooled, or recently deionized, and stored to exclude absorption of atmospheric CO_2.

6.9 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.10 Class A glassware. Class A glassware should meet the requirements of ASTM E694 "Standard Specification for Laboratory Glass Volumetric Apparatus" and ASTM E542 "Standard Practice for Calibration of Laboratory Volumetric Apparatus."

6.11 USP Section 741. Information on melting point apparatus I or II may be found in the US Pharmacopeia standards publication USP 25-NF 20, or online at http://www.uspbpep.com/usp31/v31261/usp31nf26s1_c741.asp.

6.12 Changes incorporated. This specification was updated from MIL-M-19719A to MIL-DTL-19719B as part of the Munitions Chemical Specification Modernization (MCSM) program. All test procedures were verified and updated as needed to produce valid results.

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Reference documents have been verified, updated, and hyperlinked to sources. Sampling procedures have been updated.

Specification ID	ECP / ERR
MIL-DTL-19719B	Y24Q2005

6.13 Subject term (key word) listing.

Propellant
Deterrent
Stabilizer
Munitions Chemical Specification Modernization (MCSM)

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – OS

Preparing activity:
Army – AR
Project No. 6810-2024-017

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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